DRAFT v2 section (for review; not for deposit): real related languages written in different texts

Status: draft written 2026-10-20, revised 2026-10-21 after an audit for oversights, 2026-10-24 with Finnish-Estonian (entry 9) and 2026-10-27 for entry 10 (no-source gold rows read as unknown; the synthetic sisters reframed as optimistic upper references), on branch `v2`, from the reports named in each paragraph. It does not change any v1 number or the v1 paper (`paper-v1`). Every number here is a v2 result, and every decimal in it is checked against a generated report (`reports/v2/cd/analysis/check_numbers.py`). Methods are frozen at `frozen-v2` (commit `4ce3e90`); pre-registration entries 1-10 are in `docs/v2_preregistration.md`; entry 10 changed only the v2 scoring and yardstick harness. Each claim is labelled pre-registered, exploratory or post hoc. Nothing here is, or claims, a decipherment.

Key findings

- **0 of 12** cells (3 real relatives × 4 script types) reach a 50% mean, on either seed block. Pre-registered (entry 6) and replicated on fresh seeds (entry 7).
- **2 of 12** cells show the relative beating an unrelated language by the rule fixed before that control ran: Kannada, alphabetic (+15.1 points) and Malayalam, syllabic (+7.6). Italian never does. Entry 8, C1.
- **14.0-19.7%** of alphabetic tokens are recovered with an unrelated Finnish reference, scored against the relatives' own gold, and a constant answer gets 14.2-15.8%. An unrelated reference thus reaches more than half of the 20.3-34.4% that a real relative reaches (the shares, not necessarily the same tokens). Entry 8, C1 and C2.
- **+11.5 points** for Old Tamil-Malayalam, alphabetic, with 10 EM restarts instead of 3 (30.4% to 41.9%): the only first-order budget effect. All 120 reruns at 3 restarts reproduce the frozen records exactly. Entry 8, C3 and C4.
- **Synthetic sisters are optimistic, not bounds.** Italian scores below the synthetic `harsh` setting on both seed blocks, although it is closer to Latin on every yardstick, and more restarts widen the gap (`harsh` 43.6%, Italian 18.1%); Estonian falls below Finnish `harsh` too. Real relatives are therefore the realistic `related` tier, and the synthetic sisters (v1 sister and `harsh`) optimistic upper references. Observation post hoc (negative result 3); framing, entry 10.
- **+20.1 points** for Finnish-Estonian (sister), alphabetic, over the better unrelated reference, ahead in 11 of 12 paired corpora (31.1% pooled): the clearest case of a relative helping. It still falls below `harsh` (35.9% on seeds 0-2, 41.2% pooled), and on syllabic and logo-syllabic scripts it is at the constant-answer floor. Its gold is weaker than the others': 13 of 27 rows, built from the word pairs Pajusalu (2012) prints. Entry 9; reported separately.
- **0.0%** logographic recovery with every real relative, below a constant answer (8.6-17.5%). The solver can only output the reference's 250 most frequent words, and the hidden language's frequent words have rare cognates: Latin *et* is rank 2,592 in Dante. Exploratory.

Epistemic status of the main claims

Claim	Evidence	Status	Robustness	What would change it
No real relative brings a script type to a 50% mean	Tables v2.3, v2.5; R1	Pre-registered (entries 6, 7)	Replicated; holds at 10 restarts (highest 41.9%)	A stronger solver; other relatives or texts
A relative helps measurably in only 2 of 12 cells	C1	Procedure pre-registered (entry 8), motivation post hoc	One run of 12 corpora per cell; the rule's thresholds are a choice	Other unrelated references; other gold tables
Old Tamil alphabetic recovery is in the range of the Ugaritic anchor	R2; Ugaritic report	Pre-registered check	Replicated	Nothing here: it is a consistency check, not confirmation
Real relatives fall below the synthetic `harsh` reference, so synthetic sisters are optimistic upper references	Tables v2.3, v2.5, v2.6; R3; C3; E3	Observation post hoc; R3 and E3 pre-registered; framing post hoc (entry 10)	Replicated for Italian; widens with more restarts; seen for Estonian	A genre-matched relative text; other relatives
Closeness on the yardsticks does not predict Task D	Figure v2.3; frequency check	Exploratory	Five points	More pairs
Logographic zeros are partly a solver artefact	Reachability check	Exploratory	Computed exactly, no solver	A solver without a unit cap
Solver budget is second-order except for one cell	C3	Pre-registered	Seeds 0-2 only	More restarts or another optimiser
The runs are deterministic	C4	Pre-registered	120 of 120 identical	-
Estonian measurably helps Finnish on alphabetic scripts (+20.1; 11 of 12)	Entry 9, E2	Pre-registered (entry 9)	One pair; 12 corpora; **weak gold: 10 word pairs, 13 of 27 letters**	A fuller Finnish-Estonian gold source; another Estonian text

v2.1 Why real relatives

The v1 `related` tier used a synthetic sister cut from the other half of the hidden language's own source text. It shares phrases with the hidden text and keeps its vocabulary, morphology and word order, so it is far closer than any attested relative of an undeciphered script is likely to be (v1, Section 3.6, Table 2). Stage 2 first tried to make the synthetic sister realistic by calibrating it to the distance between Ugaritic and Hebrew, the classic real case for computational decipherment (Snyder, Barzilay & Knight 2010). That failed for all five languages in both settings tried: every mechanism that changes letter pairs also destroys shared word forms, and a sister cut from the same text keeps too much verbatim text (`docs/v2_negative_results.md`, item 2). Stage 2 therefore measures real related languages written in different texts, with no tuning and no targets: whatever distance each real pair has is the distance used (pre-registration entry 5).

Entry 4 had planned to bracket the real case between two synthetic sisters: the v1 sister as an upper bound and a harsher sister-v4 setting, `harsh`, as a lower bound. That framing

is dropped (entry 10). Real relatives fall below `harsh` in two pairs on alphabetic scripts, so `harsh` is no lower bound. Both synthetic sisters are built from the hidden language's own source text and keep its vocabulary and genre, so they are reported as **optimistic upper references**. The real relatives are the realistic `related` tier and the primary evidence.

v2.2 Pairs, texts and gold correspondences

Table v2.1. Real relatives. Each pair is labelled by relationship in every table.

Hidden (v1 source)	Relative	Relationship	Relative's text	Gold correspondences (cited per row)	Status
Latin (Caesar, Vergil)	Italian	descendant, of spoken (Vulgar) Latin rather than the Classical written language (Grandgent 1907: §§3-4)	Dante, *La Divina Commedia*, Project Gutenberg #1000 (public domain in the USA)	Grandgent (1907), section per row	measured
Old Tamil (Sangam anthologies)	Malayalam	descendant: "a very ancient offshoot of Tamil… not a sister of Tamil, but a daughter" (Caldwell 1875: 23-24; Gundert, reported there, disagreed)	Ezhuthachan, *Adhyātma Rāmāyaṇam Kiḷippāṭṭu*, Malayalam Wikisource (CC BY-SA 4.0)	Caldwell (1875), pages per row	measured
Old Tamil	Kannada	sister: one of Tamil's "sister idioms" (Caldwell 1875: vi), a separate cultivated language (33-34)	Kumāravyāsa, *Karṇāṭa Bhārata Kathāmañjari*, Kannada Wikisource (CC BY-SA 4.0)	Caldwell (1875), pages per row	measured
Finnish (Kalevala)	Estonian (**weak gold**: 10 word pairs, 13 of 27 letters)	sister (both Finnic; Estonian is Southern Finnic, Finnish Northern: Pajusalu 2012: 202-203, 217)	Kreutzwald, *Kalevipoeg*, Estonian Wikisource, cantos I-XX (CC BY-SA 4.0 site licence; text public domain)	Pajusalu (2012), from the word pairs it prints; Kallio (2016) as cross-check	measured (entry 9)
Sanskrit (Rāmāyaṇa)	Pali	descendant	-	-	blocked: SuttaCentral asks that its texts not be used for AI datasets "or similar"; permission to be requested
Sumerian	-	isolate	-	-	`none` tier only

Data notes.

- The two Wikisource transcriptions state no source edition. They were accepted as clean by the project owner (2026-10-15): they carry the CC BY-SA site licence and do not link to the CC BY-NC-ND scan on Internet Archive.
- Kannada coverage is partial: 93 of the 154 sandhi pages linked from the ten parva pages exist, mostly Ādi, Virāṭa and Karṇa parvas. The two commentary (ಸಟೀಕಾ) pages are excluded. Each sandhi page opens with a short invocation that may be in Sanskrit; it was not removed.
- Revision ids and SHA-256 hashes per page are kept with the raw data.

Registers, periods and sizes differ across pairs, and none of this is controlled.

- *Register*: Both hidden languages are literary registers (Classical Latin prose and verse; Sangam poetry), while each relative descends from, or is related to, a spoken form. Italian continues Vulgar, not Classical, Latin (Grandgent 1907: §§3-4).
- *Period.* The relatives are written many centuries after the hidden texts: Dante (14th century) after Caesar and Vergil; Ezhuthachan (16th century) and Kumāravyāsa (15th century) after the Sangam anthologies. Kumāravyāsa's Kannada shows the change of initial *p* to *h* that Caldwell calls "comparatively recent" (1875: 53): in our text *hattu* 'ten' occurs 28 times and *pattu* once. The gold keeps *p* as the regular reflex and *h* as attested, so strict scoring is conservative for Kannada.
- *Loans.* Caldwell (1875: 23) notes that Sanskrit loans are fewest in Tamil and most numerous in Malayalam; our Old Tamil texts are Sangam poetry, with few.
- *Reference size.* Italian has 391,950 units, Malayalam 479,851 and Kannada 857,617. The v1 Latin and Tamil references, cut from the same half of the source as their synthetic sisters, have 118,647 and 502,785 alphabetic units (`reports/v2/ugaritic/results.md`). Reference size may affect the solver; it is not controlled.

Units and spelling. Units are those of the v1 hidden language: Latin letters by the v1 greedy segmentation, and Old Tamil phonemes. Malayalam and Kannada script are converted to the same phonemes by deterministic tables, tested on hand-checked words. Every pair is measured as written and normalised, and the normalised value is primary, so that spelling conventions do not count as language distance (the Hebrew lesson of v2 entry 3):

- Italian: accents stripped, j folded to i, silent h dropped, apostrophes split words.
- Malayalam and Kannada: the voicing and aspiration that Tamil script cannot write are collapsed (kh, g, gh → k, and so on).
- Estonian: letters as units; b, d, g written p, t, k, since in native words they spell short lenis stops (Pajusalu 2012: 213). Estonian writes a long stop after a long vowel single where Finnish writes it double (*saatma*, Finnish *saattaa*); this is left as written, which can only make the pair look less alike.

Gold tables. Each hidden unit has a regular reflex (its outcome where no conditioned change applies) and a list of attested reflexes, with the section or page of the source for every row. Rows the source does not cover are marked "no source" and are unrecoverable under strict scoring (Old Tamil h, s, k̲, ṣ for both Dravidian relatives). Since entry 10 a no-

source row is read as an unknown reflex everywhere: a syllable or word containing it is unrecoverable, and the yardsticks break the text there. Before entry 10 the harness dropped such a letter like a lost sound; only Latin *h*, which Grandgent states is lost (§249), is now treated that way. Both sources are over a century old; their basic consonant correspondences are standard, but details are dated. No gold value comes from Wikipedia.

The Finnish-Estonian gold is built differently, because its named source, Pajusalu (2012), describes the innovations of the Southern Finnic languages and has no correspondence table. The table uses only the ten Finnish-North Estonian word pairs that paper prints (for example *sisar ~ sõsar*, *kato ~ kadu*, *kolmas ~ kolmas*; pp. 208-218), aligned letter by letter in the code. A letter kept in a pair gives the regular reflex; a change the paper describes as an innovation (õ for *a, e, i, o*; non-initial *o* > *u*) is attested only. That gives a regular reflex for 13 of 27 rows. Finnish *p, v, j, y, ö* and *ä* have no kept example and are "no source", although *y* and Estonian *ü* spell the same sound: no general knowledge was used. The rule was fixed in entry 9 before any measurement.

On 2026-10-21 four rows were spot-checked against the scanned texts, and each citation holds: Kannada initial *p* > *h* (Caldwell 1875: 53), *talei* = Malayalam *tala* (p. 19), Malayalam as an offshoot of Tamil (pp. 23-24) and Latin *x* > *s* (Grandgent 1907: §255). This is a spot check, not a full audit.

v2.3 How far the real relatives are

The Ugaritic-Hebrew yardsticks (v2 anchor) are computed with the same code on each pair. Sample sizes are matched to the anchor, and chance is estimated from shuffled correspondence maps.

Cross-script comparisons are indicative only. Shared word forms and word pairs are not directly comparable between a consonant-only script (Ugaritic-Hebrew) and scripts that write vowels (all the other rows): a consonant skeleton matches far more easily than a full word, even after the chance correction. Comparisons within one script type are the meaningful ones.

Table v2.2. Yardsticks, normalised spelling (primary). cc = chance-corrected. Real-pair intervals: bootstrap over the relative's lines, 200 replicates, 95%. No-source gold rows read as unknown (entry 10).

Pair	Relationship	Script type	JSD above floor (bits)	Shared word forms, cc	Word pairs, cc	Verbatim 6+ units
Ugaritic-Hebrew (vowel letters removed)	sister	consonant-only	0.089	58.4%	4.6%	0.8%
Latin-Italian	descendant	alphabet	0.104 (0.101-0.108)	11.8% (9.6-11.5%)	0.6% (0.1-0.7%)	1.9% (0.7-3.3%)
Old Tamil-Malayalam	descendant	abugida	0.141 (0.137-0.144)	5.9% (4.2-5.3%)	0.0% (0.0-0.0%)	0.8% (0.6-2.2%)
Old Tamil-Kannada	sister	abugida	0.181 (0.178-0.186)	5.8% (4.5-5.6%)	0.0% (0.0-0.1%)	1.2% (0.4-2.4%)
Finnish-Estonian† (weak gold: 10 word pairs, 13 of 27 letters)	sister	alphabet	0.183 (0.178-0.186)	10.6% (8.9-10.7%)	0.6% (0.4-0.9%)	1.3% (1.0-4.4%)
Finnish, v1-like sister* / `harsh`	synthetic, optimistic	as hidden	0.004 / 0.154	53.9% / 4.8%	10.4% / 0.2%	19.8% / 3.5%
Latin, v1-like sister* / `harsh`	synthetic, optimistic	as hidden	0.006 / 0.135	53.1% / 7.8%	7.1% / 1.1%	12.4% / 2.4%
Old Tamil, v1-like sister* / `harsh`	synthetic, optimistic	as hidden	0.002 / 0.185	57.6% / 3.6%	8.5% / 0.1%	17.8% / 1.8%

* The yardstick stand-in for the v1 sister: sister-v4 with v1-like settings (`reports/v2/sister/`), not the v1 sister object used in the Task D tables. † Weak gold (entry 9): 14 of 27 Finnish letters have no source, 12.6% of the hidden letters. Before entry 10 the yardstick code deleted them, which cut mapped words short and made the word-level yardsticks negative (-7.9% shared word forms; `reports/v2/cd/entry10.md`). Read as unknown, words containing them are left out, and Finnish-Estonian shares about as many word forms as Latin-Italian. The letter-pair distance stays high, because the Estonian text still contains the letters the Finnish side now leaves out.

For Latin-Italian, Malayalam and Kannada the plug-in estimate of shared word forms lies above its own bootstrap interval, because resampling duplicates lines and so lowers the share of distinct forms; the interval is the conservative reading.

Reading.

- On these yardsticks the real relatives sit near the `harsh` setting and far from the v1-like sister, which shares about 4-10 times as many word forms.
- Malayalam is closer to Old Tamil than Kannada on the letter-pair distance, and the intervals do not overlap. That fits the relationship labels, but the two texts also differ in genre, period and size, so the distance cannot be attributed to the relationship alone.
- Normalisation matters most where scripts differ most. Kannada writes voicing; collapsing it moves the letter-pair distance from 0.310 to 0.181 bits.

What the verbatim matches are (exploratory inspection, `reports/v2/real_relatives/verbatim.md`). Verbatim runs of 6+ units occur in 1.6% (Malayalam), 1.5% (Kannada) and 2.5% (Italian) of sampled windows, against 0.0-0.1%

through shuffled maps and 0.3-0.4% for an unrelated language with a similar open syllable shape (the v1 Finnish hidden half, letters as they are). They are therefore signal, but not shared text:

- 60-74% of them run across a word boundary;
- they are mostly short shared morphs and phoneme strings (for example Malayalam *-kkoṇṭu*, *colliyatu*, *paricil*), and for Italian, cognate stems and Dante's Latinisms (*principi-*, *cantando*, *crudel-*);
- no match is a phrase that respects word boundaries on both sides, so they are not shared formulae.

For Finnish-Estonian the share is 3.1% (shuffled maps: 0.4% at the 95th percentile; unrelated Latin letters: 0.3%), and no match spans two whole words. Under entry 10 a window cannot cross a letter without a source, so the windows come from stretches written in letters the gold covers, which raises the share (1.2% before entry 10). Kreutzwald modelled *Kalevipoeg* on the *Kalevala*, so shared formulas were possible; none is found at this level.

In Malayalam about a fifth of the matches (33 of 164) exist only after normalisation. By inspection, most of these pair a native Tamil string with a Sanskrit loanword: the loan is a coincidental partner, not a shared loan.

v2.4 Tasks C and D with real relatives

Design (entries 1, 2, 4-6).

- **Corpora:** the v1 Indus-point corpora (2,906 texts, mean length 4.6 signs), seeds 0-2, samplers A and B, four script types, with the corrected (content-split) sister: 6 corpora per hidden language and script type. They were regenerated from the same seeds; all 120 have corpus statistics identical to the v1 records (checked 2026-10-20).
- **References:** the relative's whole text at the script's unit level, as v1 writes its references:
  - phonemes for alphabetic scripts;
  - CV syllables for syllabic scripts;
  - word forms for logographic scripts;
  - for logo-syllabic scripts, word forms for the relative's 250 most frequent words and syllables otherwise.

  No reference carries a correspondence table, so every score is before the cognate step.
- **Tiers:** `related` gives the relative alone. `candidates` gives the relative plus the other v1 languages.
- **Methods:** the frozen EM solver after Knight et al. (2006), with the revised selection rule (primary) and the original rule, and the frequency-rank baseline. Each EM fit keeps the best of 3 restarts: a frequency-informed start (sign rank k to unit rank k) and two random starts.
- **Scoring:**

    - Strict (primary): a sign token is right when its predicted unit is the regular reflex of the token's hidden value. A token whose hidden value contains a no-source letter is unrecoverable (entry 10). A logogram is right only when its reflex is a word that occurs in the relative (a cognate). Many-to-one predictions are accepted.
    - Lenient (secondary): any combination of attested reflexes counts.
    - Every token counts. The share of tokens that have a strict reflex at all is reported as a ceiling.
- **Synthetic reference rows:** beside the real relatives stand the v1 sister (read from the v1 records and scored with the v1 scorer before the cognate step) and `harsh`. Both are optimistic upper references (entry 10); entry 4 had labelled them upper and lower bound, and `harsh` is not a lower bound (negative result 3).

Table v2.3. Task D, `related` tier, revised EM, strict score: mean % of sign tokens over 6 corpora (number of corpora at or above 50%). From `reports/v2/cd/results.md`.

Hidden	Relative	Relationship	Logographic	Syllabic	Logo-syllabic	Alphabetic
Old Tamil	v1 sister	synthetic, optimistic	0.0 (0)	26.7 (0)	19.0 (0)	42.5 (2)
Old Tamil	**Malayalam**	descendant	0.0 (0)	6.2 (0)	5.9 (0)	**30.4 (1)**
Old Tamil	**Kannada**	sister	0.0 (0)	3.3 (0)	2.9 (0)	**32.2 (0)**
Old Tamil	`harsh`	synthetic, optimistic	0.1 (0)	6.2 (0)	5.9 (0)	16.4 (0)
Latin	v1 sister	synthetic, optimistic	1.4 (0)	32.2 (1)	9.8 (0)	45.4 (2)
Latin	**Italian**	descendant	0.0 (0)	6.0 (0)	1.5 (0)	**16.9 (1)**
Latin	`harsh`	synthetic, optimistic	1.5 (0)	24.4 (0)	4.6 (0)	36.1 (1)

The real Ugaritic anchor, for comparison (`reports/v2/ugaritic/results.md`; pre-registered 50% criterion failed): 32.0% (revised EM) and 33.9% (original EM) on natural texts, `related` tier, alphabetic. The like-for-like condition, texts re-cut to the Indus mean length of 4.6, gives 24.6-29.9% with revised EM.

Table v2.4. Task C, `candidates` tier: share of corpora (24 per row, all script types) where the solver picked the right family, and the references the revised rule chose.

Hidden	Relative offered	EM revised	EM original	Frequency rank	Chosen by EM revised
Old Tamil	Malayalam	0%	0%	38%	Latin 16, Finnish 8
Old Tamil	Kannada	38%	0%	8%	Latin 10, Kannada 9, Finnish 5
Latin	Italian	25%	0%	50%	Finnish 18, Italian 6
Old Tamil	v1 sister	33%	13%	13%	Latin 13, sister 8, Finnish 3
Latin	v1 sister	96%	8%	63%	sister 23, Finnish 1

For Latin, picking Sanskrit also counts as the right family (Indo-European), as in v1.

Fresh-seed replication (entry 7, pre-registered). The whole run was repeated on seeds 3-5 (120 corpora, no errors; the corpora were checked identical to the v1 records

before any score was read). All four checks fixed before the run replicated (`reports/v2/cd/replication.md`):

- R1: every real-relative mean stays below 50% (highest: Kannada, alphabetic, 36.7%).
- R2: Old Tamil alphabetic stays within 10 points of Ugaritic: Malayalam 30.5%, Kannada 36.7%.
- R3: Italian stays below `harsh` for alphabetic (23.8% against 26.8%) and syllabic scripts (2.4% against 9.4%). The alphabetic margin is small.
- R4: the revised rule picks the real relative in 0 (Malayalam), 10 (Kannada) and 6 (Italian) of 24 corpora.

Some cells move a lot between the two seed blocks (Malayalam syllabic 6.2% then 15.1%; Old Tamil `harsh` alphabetic 16.4% then 9.5%). The pooled means over all 12 corpora are therefore the steadier summary, and they are given here beside the primary seeds 0-2 tables.

Table v2.5. Task D, `related` tier, revised EM, strict: mean over 12 corpora (seeds 0-5) with a 95% bootstrap interval over corpora (descriptive; `reports/v2/cd/pooled.md`).

Hidden	Relative	Relationship	Syllabic	Logo-syllabic	Alphabetic
Old Tamil	v1 sister	synthetic, optimistic	29.3 (21.9-38.1)	21.3 (11.1-32.3)	43.5 (34.4-53.4)
Old Tamil	Malayalam	descendant	10.7 (6.9-15.7)	6.6 (3.9-9.7)	30.5 (16.9-44.2)
Old Tamil	Kannada	sister	4.0 (2.5-5.8)	3.9 (2.0-6.2)	34.4 (28.8-41.2)
Old Tamil	`harsh`	synthetic, optimistic	8.1 (4.3-12.4)	6.3 (3.0-9.8)	12.9 (7.0-19.2)
Latin	v1 sister	synthetic, optimistic	29.3 (16.2-43.1)	9.7 (5.8-14.4)	46.3 (37.6-55.3)
Latin	Italian	descendant	4.2 (1.8-6.9)	2.3 (1.2-3.9)	20.3 (9.6-32.6)
Latin	`harsh`	synthetic, optimistic	16.9 (9.3-25.4)	3.6 (1.8-5.6)	31.5 (24.0-38.9)

Logographic scripts are left out of this table: every real relative scores 0.0 there (the full table is in `pooled.md`).

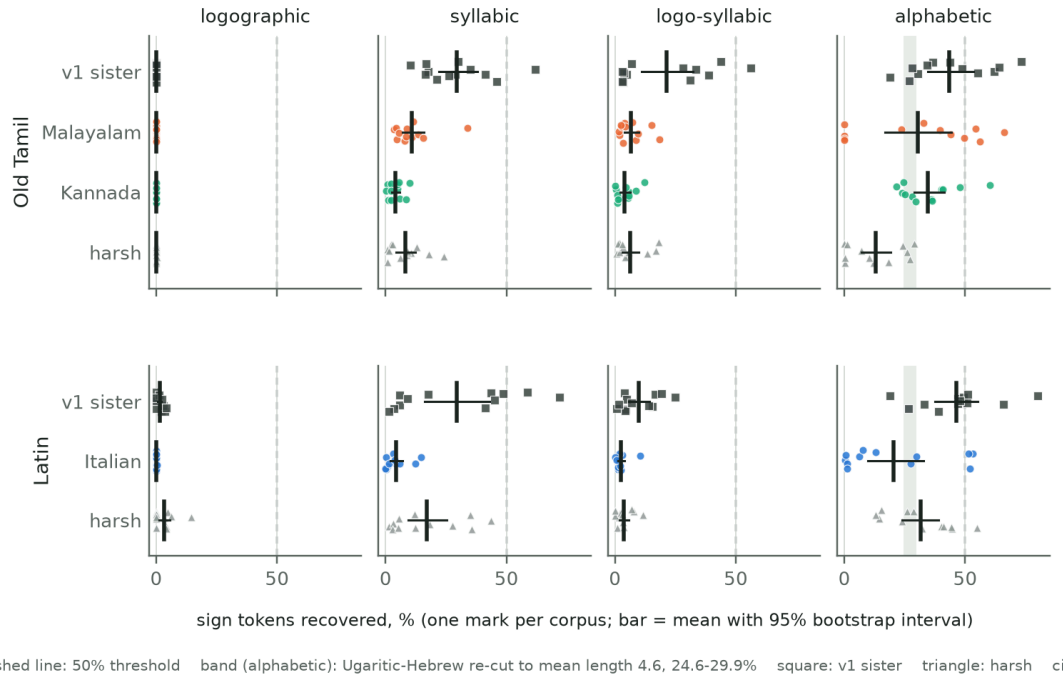


Figure v2.1. Task D per corpus, `related` tier, revised EM, strict, seeds 0-5 (12 corpora per row). Bar: mean with 95% bootstrap interval. Band (alphabetic): the Ugaritic anchor re-cut to mean length 4.6. Scores are often near 0% or near 50% for the same relative and script type.

Finnish-Estonian (entry 9, pre-registered; `reports/v2/cd/estonian.md`). The same harness on Finnish with *Kalevipoeg*, run after the yardsticks were reported: 24 corpora on seeds 0-2 (primary), 24 on seeds 3-5 and 48 for the controls, all on the frozen code, with no errors. Its gold has a regular reflex for 13 of 27 rows, so it is reported in its own table and not pooled with the other pairs.

Table v2.6. Finnish-Estonian (weak gold: 10 word pairs, 13 of 27 letters), Task D, `related` tier, revised EM, strict (%). Pooled: seeds 0-5 with a 95% bootstrap interval over corpora.

Script	Estonian, seeds 0-2	Pooled 0-5	Ceiling	`harsh` (synthetic, optimistic), seeds 0-2	v1 sister (synthetic, optimistic), seeds 0-2	Task C: picks Estonian (of 6)
Logographic	0.0	0.2 (0.0-0.5)	15.5	0.0	1.0	0
Syllabic	9.3	8.1 (4.4-12.3)	82.9	19.9	43.6	6
Logo-syllabic	3.7	3.6 (2.1-5.1)	50.5	8.6	21.2	6
Alphabetic	29.8	31.1 (25.1-36.3)	86.5	35.9	65.3	3

- No Estonian run reaches 50%; the highest is 44.1% (alphabetic). Alphabetic runs range from 5.3% to 44.1%.
- Estonian falls below `harsh` in every script type but logographic (E3), as Italian does on alphabetic and syllabic scripts: a real sister pair also scores below the optimistic synthetic reference.

- Entry 10 lowered the ceilings (syllabic 96.5% to 82.9%, logo-syllabic 60.5% to 50.5%, logographic 22.4% to 15.5%): before it, a syllable containing a no-source consonant counted as recoverable with the consonant dropped. Scores moved by at most 0.5 points, and no reading flipped.
- In `candidates_real`, the revised rule picks Estonian in all 12 syllabic and logo-syllabic corpora, in 3 of 6 alphabetic ones (Latin in the rest) and in none of the logographic ones (E4: 15 of 24).

v2.5 Controls: is the relative doing the work?

Pre-registration entry 8 fixed four controls and their readings before they ran (post hoc in motivation). Results in `reports/v2/cd/controls.md`.

C1. Unrelated reference. The same corpora are run with an unrelated v1 language as the only reference, and its predictions are scored against the real relative's gold, exactly as the relative's own run. A relative *helps* in a cell only if it adds at least 5 points over the better unrelated reference and is ahead in at least 9 of the 12 paired corpora.

- It helps in 2 of 12 cells: Kannada, alphabetic (34.4% against Finnish 19.4%; +15.1; 9 of 12) and Malayalam, syllabic (10.7% against Latin 3.1%; +7.6; 10 of 12).
- Malayalam alphabetic adds 10.7 points but is ahead in only 7 of 12 corpora, because several of its runs fail outright.
- Italian helps nowhere: +6.3 on alphabetic (6 of 12), and below Finnish on syllabic (-1.2) and logo-syllabic scripts (-0.2).
- Finnish, which is unrelated to both hidden languages, recovers 14.0-19.7% of alphabetic tokens against the relatives' gold. Its transcription shares many letters with ours (*a, i, u, k, t, p, n*), so coincidence alone reaches that level.
- Estonian (entry 9, same rule, Latin and Old Tamil as the unrelated references) helps on alphabetic scripts: 31.1% against Latin 11.1%, +20.1 points, ahead in 11 of 12 paired corpora. It does not help on syllabic (+5.5; 7 of 12), logo-syllabic (+2.5; 10 of 12) or logographic scripts (+0.2).

C2. Constant-answer floor. Mapping every sign to the single most common regular reflex scores 14.2-15.8% on alphabetic scripts. The real relatives stay above it there. On logographic scripts the floor is 8.6-17.5% and every real relative scores 0.0%; on syllabic and logo-syllabic scripts, Kannada and Italian are at or below it. For Estonian the floor is 12.9% (alphabetic), 12.6% (syllabic), 9.3% (logo-syllabic) and 5.8% (logographic): Estonian is above it only on alphabetic scripts.

C3. Solver budget. With 10 EM restarts instead of 3 (the first three identical), only Old Tamil-Malayalam, alphabetic, moves by more than 10 points: 30.4% to 41.9%. Kannada alphabetic falls from 32.2% to 28.9%, Italian alphabetic rises from 16.9% to 18.1%, and Latin `harsh` alphabetic rises from 36.1% to 43.6%. A higher likelihood does not always mean more signs right.

C4. Determinism. The 3-restart reruns reproduce all 120 frozen-run values exactly.

Degeneracy. The share of sign tokens assigned to the single most common predicted unit is 12.4% (Old Tamil) and 12.2% (Latin) with the real relatives, against 11.1% and 11.2% with unrelated references: the solutions do not collapse onto one unit.

Reachability (exploratory, `reports/v2/cd/reach_check.md`). The entry-2 ceiling counts a token as recoverable if its reflex occurs anywhere in the relative, but the frozen solver only predicts among the reference's 250 most frequent units. On logographic scripts the reachable share is far lower than the ceiling: 25.4% against 51.4% (Italian), 2.3% against 24.3% (Kannada) and 10.1% against 16.9% (Malayalam). On syllabic and alphabetic scripts the cap hardly binds. The v1 sister, cut from the same text as the hidden corpus, shares its frequent words, so the cap rarely binds for it. Logographic comparisons between the synthetic sister and real relatives are therefore partly a solver artefact.

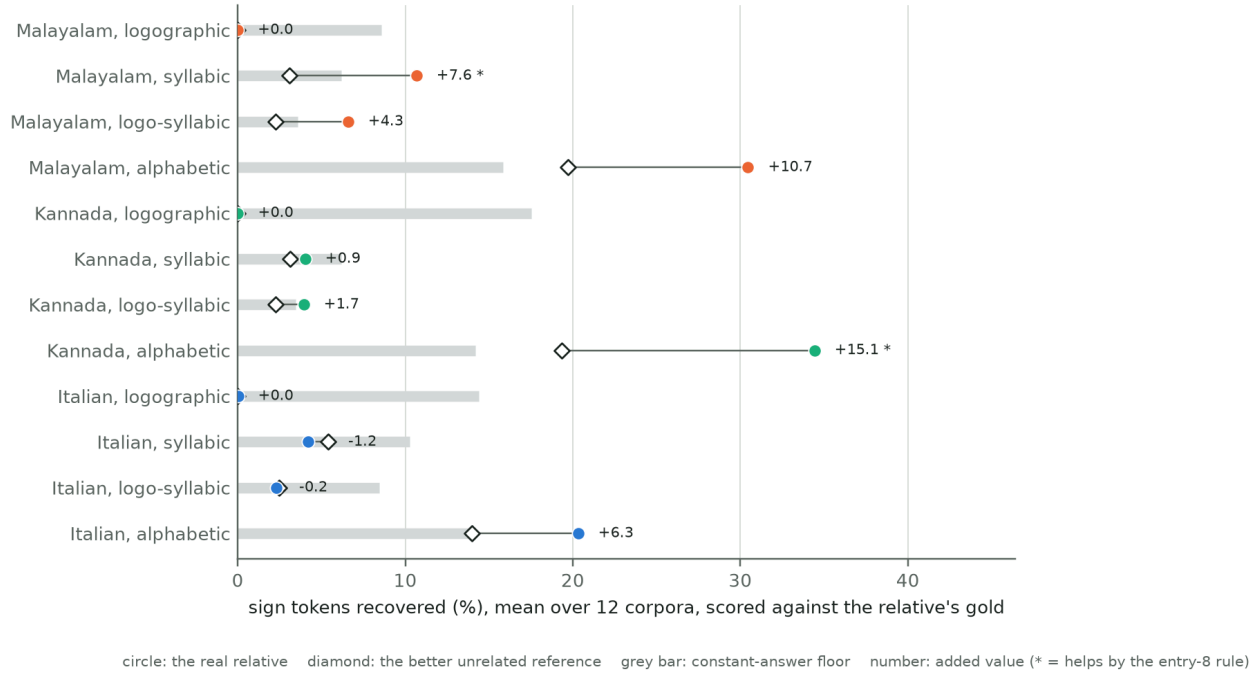


*Figure v2.2. Each real relative (circle) against the better unrelated reference (diamond) and the constant-answer floor (grey bar), mean over 12 corpora, scored against the relative's gold. Number: added value; * = helps by the entry-8 rule.*

v2.6 What the results show

1. **No real relative brings any script type to a 50% mean** (pre-registered, replicated).
    - In the `related` tier, 7 of the 144 real-relative runs reach 50% (3 relatives, 4 script types, 12 corpora each): 2 on seeds 0-2 and 5 on seeds 3-5, all alphabetic. Single runs reach 66.0% (Malayalam), while every cell mean stays at or below 36.7% (`reports/v2/cd/pooled.md`).
    - With 10 restarts the highest cell mean on seeds 0-2 is 41.9% (Malayalam, alphabetic).
    - Syllabic and logo-syllabic scripts reach 2.3-10.7% with a real relative over 12 corpora; logographic scripts reach 0.0%.
2. **Most of the alphabetic recovery does not need the relative.** An unrelated Finnish reference recovers 14.0-19.7% against the relatives' gold, and a constant answer 14.2-15.8%. The relatives add 6.3-15.1 points over the better unrelated reference on alphabetic scripts, and by the pre-registered rule help measurably in 2 of 12 cells. Italian helps in none. Finnish-Estonian is the exception on alphabetic

scripts (+20.1 points, 11 of 12; entry 9), the largest added value of any pair, with a gold that covers fewer letters.

3. **The Old Tamil alphabetic scores are in the range of the Ugaritic anchor** (30.5% and 34.4% against 24.6-29.9% like-for-like), as pre-registered check R2 required. This is a consistency check, not corroboration: the conditions differ (section v2.7).
4. **Real relatives fall below the synthetic references** (post hoc; `docs/v2_negative_results.md`, item 3). The synthetic sisters are therefore optimistic upper references, not bounds (entry 10).
    - Italian falls below `harsh` on both seed blocks for alphabetic and syllabic scripts (pooled alphabetic 20.3% against 31.5%), and more restarts widen the gap.
    - Kannada falls below `harsh` for syllabic and logo-syllabic scripts on both seed blocks.
    - Estonian, a sister of Finnish, falls below `harsh` for syllabic, logo-syllabic and alphabetic scripts (alphabetic 29.8% against 35.9% on seeds 0-2; entry 9, E3). This is a fourth pair with the same pattern, so the failure is not specific to a descendant relationship.
    - The real relatives are therefore the primary evidence for this tier, and the bracket of entry 4 is dropped. These reporting decisions were taken after seeing the results.
5. **Closeness on the yardsticks did not predict Task D** (exploratory; Figure v2.3). Italian is closer to Latin than `harsh` on every yardstick and agrees better in letter frequencies (unigram JSD 0.026 against 0.059; `reports/v2/cd/frequency_check.md`), yet scores lower. A text effect of the needed size is documented in the Ugaritic anchor, where a genre-matched Hebrew reference added 5-16 points.
6. **Logographic zeros are partly a solver artefact** (exploratory; section v2.5). The solver's 250-unit cap excludes most cognates of the hidden language's frequent words.
7. **Task C rarely finds the real relative** (pre-registered check R4, replicated). The revised rule picks it in 0 (Malayalam), 9 and 10 (Kannada) and 6 and 6 (Italian) of 24 corpora on the two seed blocks, mostly preferring Latin or Finnish; the original rule picks Sumerian in all 72 corpora of seeds 0-2. Estonian is picked in 15 of 24 corpora (entry 9, E4), the most of any real relative.
8. **Single runs are unstable, and budget matters for one cell.** Scores for the same relative and script type range from 0.0% to 66.0% between the 12 corpora (Malayalam, alphabetic), and 10 restarts raise Malayalam alphabetic by 11.5 points.

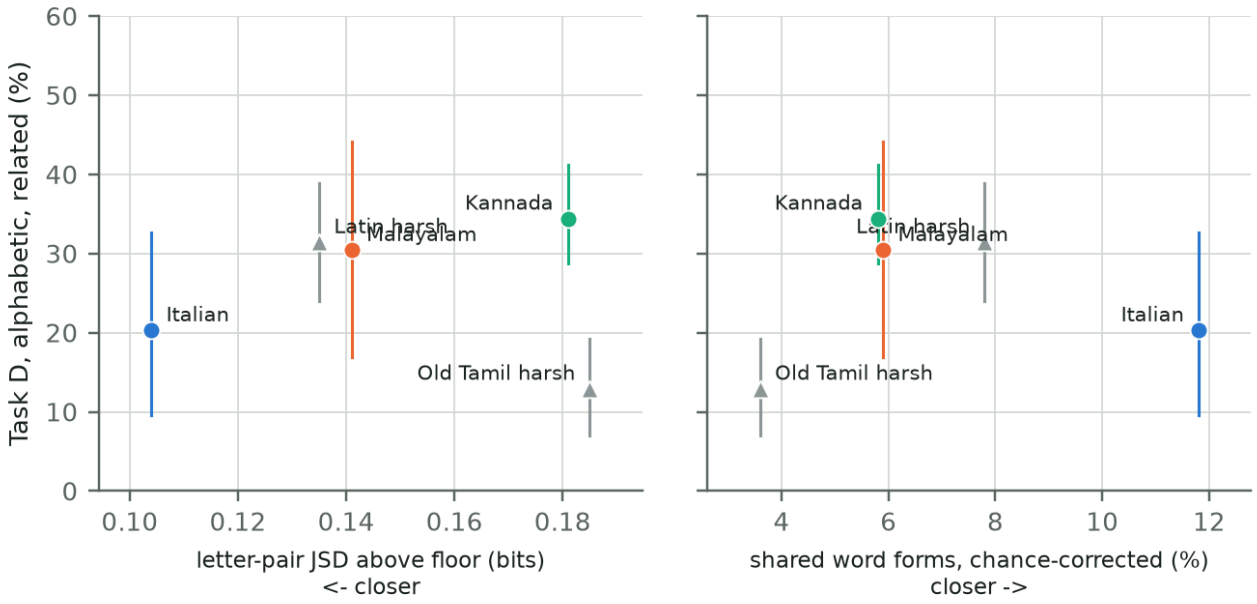


Figure v2.3. Alphabetic Task D (`related` tier, mean and 95% interval over 12 corpora) against two yardsticks for the same pairs (`harsh` yardsticks measured on seeds 0-2). Same-script pairs only. Exploratory: five points.

For Indus, where no relative is agreed and the `candidates` and `none` tiers are the relevant ones, these results add one point. With these comparison-based solvers, a genuine relative adds little beyond what an unrelated language or a constant answer already gives, outside alphabetic scripts. Inside them the added value is modest and unstable.

v2.7 What these results do not show

- **How hard Indus is.** The hidden corpora are synthetic, generated at Indus scale from known languages; the solver is told the script type; and the relative is handed to it. Indus has no agreed relative, so the `candidates` and `none` tiers are the relevant ones there.
- **That a real relative is of little use to a human decipherer.** These solvers use neither cognates nor sound laws unless given the correspondence table, which is withheld: every score is before the cognate step. A human decipherer can use both; these scores say nothing about that.
- **That the relationship, rather than the text, drives the scores.** Each relative is one text of one genre and period. In the Ugaritic anchor, a pre-registered tier with a genre-matched Hebrew reference (the poetic books) raised revised-EM recovery from 32.0% to 37.2% on natural texts, and from 24.6-29.9% to 38.4-43.0% on texts re-cut to mean length 4.6 (`reports/v2/ugaritic/results.md`). Text effects of that size could account for real relatives scoring below `harsh`.
- **A ranking of sister against descendant.** There are two relatives of one hidden language, with different texts, periods and reference sizes.
- **Independent confirmation from Ugaritic.** The Old Tamil alphabetic scores and the like-for-like Ugaritic range are similar, but the conditions differ. Ugaritic has 6,928 sign tokens in 29 real letters with no synthetic allographs; our corpora have 12,904-13,401 tokens with allographs, homophony and polyvalence (`reports/v2/cd/pooled.md`). Hebrew letters also cannot match an unrelated reference

by coincidence, while Latin-alphabet transcriptions can (control C1). The agreement is a consistency check, not corroboration.

- **Statistical significance.** No hypothesis test was pre-registered for these comparisons, and none is reported. Intervals are descriptive.

v2.8 Limitations of this section

1. **Three hidden languages only.** Latin, Old Tamil and Finnish; Old Tamil has both relatives, so the two Dravidian rows share their hidden side and their gold source. Sanskrit has no relative with clear terms: SuttaCentral's licensing page asks that its texts not be used for AI datasets "or similar", and permission has not been requested yet (`docs/v2_pali_permission_request.md`).
2. **Relationship and text are confounded.** Each relative is one text of one genre and period; the effect of the relationship cannot be separated from that of the text.
3. **Old gold sources.** Caldwell (1875) and Grandgent (1907) are cited per row, gaps are marked rather than filled, and some details are dated.
4. **Unverified transcriptions.** The Wikisource transcriptions state no source edition, and Kannada coverage is partial.
5. **Normalisation choices.** Normalisation is fixed in code and tested, but it is a choice, and it moves the Kannada distance a great deal.
6. **Scoring choices.**
    - Task D is scored whatever reference the solver chose, as in v1, so a wrong reference can still score by coincidence.
    - The logogram cognate rule (the reflex must occur in the relative) makes logographic ceilings low by construction.
7. **v1 assumptions still apply.** The solver is told the script type (an oracle). The Ugaritic gold table is still secondary: Wikipedia, citing Kogan (2011), which we have not read.
8. **Few corpora per cell, with high variance.** Six corpora per cell in the primary tables, twelve when pooled with the replication. Single-corpus scores are often bimodal (near 0% or near 50%), and percentile bootstrap intervals over so few corpora understate uncertainty. All intervals here are descriptive.
9. **A convenience sample of pairs.** The pairs were chosen for open texts and licences, not as a representative sample of real relationships. Nothing here generalises to "real relatives" at large.
10. **One solver family.** All solvers are frequency- or EM-based (after Knight et al. 2006), with a fixed budget. Neural models and human decipherers, who use cognates and sound laws, may do better.
11. **Analysis code outside the freeze guard.** The replication and control drivers and the analysis scripts live in `reports/v2/cd/analysis/`, which the frozen-code check does not cover. They call only frozen functions; each was committed before its run, and the commits are named in the analysis history.
12. **Gold tables spot-checked, not audited.** Four rows were rechecked against the scanned sources; the rest rest on the original reading.
13. **The entry-2 ceiling ignores the solver.** It counts a token as recoverable if its reflex occurs anywhere in the relative, while the solver can only predict among the

reference's 250 most frequent units. For logographic scripts the reported ceilings overstate what the solver could reach (section v2.5). This was found after the run and is reported, not corrected.

14. **A weak gold for Finnish-Estonian.** Its source has no correspondence table, so the gold comes from ten printed word pairs and has a regular reflex for 13 of 27 letters; every table labels it weak. Since entry 10 the other letters are unknown rather than lost, which lowers the ceilings and removes the artefact that made the word-level yardsticks negative (`reports/v2/cd/entry10.md`).
15. **Frozen syllable cutter.** It uses the hidden language's vowel set, so Estonian õ and ü are cut as consonants in the Estonian syllabic reference. Neither is the regular reflex of any Finnish letter in this gold.

v2 analysis history (dated)

- **2026-10-11:** synthetic-sister calibration killed; pre-registration entry 4 (the bracket).
- **2026-10-15:** pre-registration entry 5 (real relatives) before any download. Dravidian Wikisource texts accepted as clean by the project owner. Gold tables committed with citations before any measurement.
- **2026-10-17:**
  - Yardstick table reported before any Task C/D run.
  - Verbatim matches inspected (exploratory).
  - Pre-registration entry 6 (harness details) proposed after a pipeline check on development seed 90 only. That check found a scorer error, fixed before the freeze: Sumerian numerals and written determinatives had been counted as unrecoverable.
- **2026-10-18:** entry 6 declared; methods tagged `frozen-v2` (commit `4ce3e90`); final run of 120 corpora with no errors.
- **2026-10-20 (post hoc):**
  - Per-corpus table added.
  - The bracket failure recorded as negative result 3.
  - Real relatives made the primary `related`-tier evidence; the bracket is secondary.
- **2026-10-21:**
  - Pre-registration entry 7 (fresh-seed replication) declared and pushed (commit `37be99f`) before any corpus was generated; corpora checked identical to the v1 records before any score was read. A container restart stopped the run at 114 of 120 corpora; the resumable driver finished it. All four checks R1-R4 replicated.
  - Audit for oversights. Relationship labels sourced from Caldwell and Grandgent (the unsourced "western Old Tamil dialect" removed); the like-for-like Ugaritic range used; the genre effect in the Ugaritic anchor cited; four gold rows spot-checked; the Kannada $p > h$ shift checked in our text; register, period and reference-size caveats added; stale plan statuses dated; a number-provenance check added.

    - Exploratory checks: unit-frequency agreement (refutes one explanation offered in negative result 3) and solver reachability (finds the logographic ceiling oversight).
    - Pre-registration entry 8 (controls C1-C4) declared and pushed (commit `29de37e`) before any control ran; post hoc in motivation. Run records archived with SHA-256 sums.
- **2026-10-24:**
    - Finnish-Estonian done directly on the project owner's instruction. Pajusalu (2012) taken from the Internet Archive copy of the society's URL (which now redirects) and read in full: it has no correspondence table. Pre-registration entry 9 (gold rule, normalisation, runs, readings E1-E4, a scorer diagnostic) declared and pushed (commit `30471b0`) before any measurement.
    - Yardsticks and verbatim check reported first; the negative word-level yardsticks traced to the deletion of no-source letters (exploratory, post hoc). The entry-9 row count corrected (13 of 27 rows with a regular reflex, not 14) before any Task C/D result was read.
    - Task C/D: 96 corpora, no errors. E1 was evaluated as in `controls.md` (at the floor = no more than 2 points above it).
    - Pali: SuttaCentral's terms read; still blocked; a permission request drafted, not sent.
- **2026-10-27:**
    - Pre-registration entry 10 declared and pushed (commit `d686a5a`) before any rerun: a no-source gold row is unknown, not lost, in the v2 scoring and yardstick harness only; the bracket framing dropped (synthetic sisters are optimistic upper references); the Finnish-Estonian gold labelled weak in every table.
    - 96 corpora rerun (Finnish, Old Tamil; seeds 0-5), 0 errors; every prediction scored under the old and the new reading, and the old scores matched the archived records in all 3,348 values compared. Old Tamil scores did not change at one decimal; Finnish-Estonian ceilings fell; no reading flipped. Predictions are archived so later scoring questions need no rerun.
    - An implementation error in the entry-10 verbatim inspection (a stretch after a break was counted as a word start) was found and fixed before the results were reported.

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- Texts: Dante, *La Divina Commedia*, Project Gutenberg #1000. Ezhuthachan, *Adhyātma Rāmāyaṇam Kiḷippāṭṭu*, Malayalam Wikisource. Kumāravyāsa, *Karṇāṭa Bhārata Kathāmañjari*, Kannada Wikisource. Kreutzwald, *Kalevipoeg*, Estonian Wikisource. Finnish: *Kalevala*, Project Gutenberg #7000. Ugaritic: EUPT (Göttingen), CC BY-SA 4.0. Hebrew: Open Scriptures Hebrew Bible (WLC), CC BY 4.0 annotations, text public domain.
- Not read, cited only as the source of a secondary table: Kogan (2011), via Wikipedia, "Ugaritic alphabet".